AL CHEM 2016 PAPER 2

Section A: physical and general chemistry

1.

a.

(i) Define the second ionizing energy of fluorine

(ii) Write an equation to represent the second ionization of fluorine

(iii) Using the axis below, sketch a graph to show the successive ionization energies of fluorine

(iv) Give reasons for the shape of your graph

b.

(i) Draw the "dot-and-cross" diagram for carbonic acid, H_2CO_3

(ii) Draw the resonance structures of the HCO_3 anion

(iii) Complete the table below by stating the molecular shapes and bond angles of the following species

Species	Molecular shape	Bond angle
CF_4		
NF_3		
CH_2Cl_2		
H_3O^+		

c. The following reaction occurs at 450°C

$$H_{2(g)} + I_{2(g)} = 2HI_{(g)}$$

Experiment	Initial $[H_2]$ mol dm^{-3}	Initial $[I_2]$ mol dm^{-3}	Initial rate of production of HI mol dm^{-3} s^{-1}
1	0.0113	0.0011	1.9×10^{-23}
2	0.0220	0.0033	1.1×10^{-22}
3	0.0550	0.0011	9.3×10^{-23}
4	0.0220	0.0056	1.9×10^{-22}

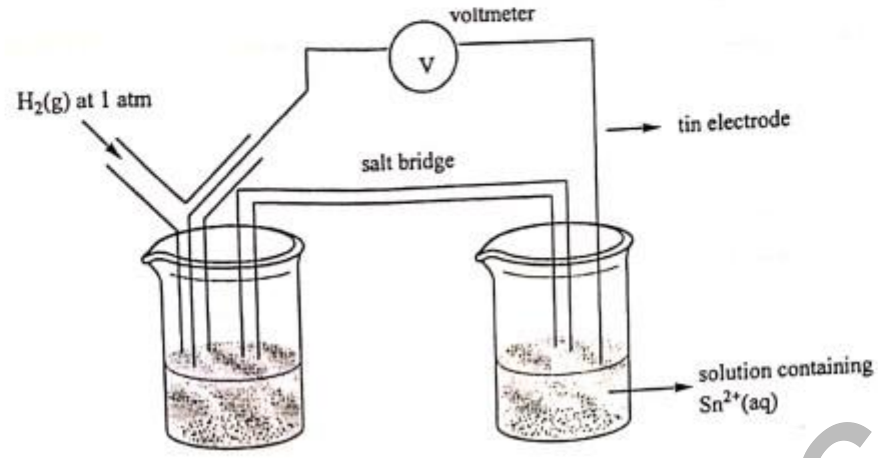
i. Determine the order of the reaction with respect to H_2 and I_2

A: Order with respect to H_2

B: Order with respect to I_2

ii. What is the overall of the reaction?

d. A student measuring standard electrode potentials has the following set up in which he connected a half-cell to standard hydrogen electrode



i. Name the solution which could be used in the left-hand beaker
ii. Suggest a solution that is suitable for the salt bridge
iii. Write the cell diagram for the arrangement above

SECTION B: INORGANIC CHEMISTRY

2.

a.

i. Write the outer electron configuration of the halogens
ii. State and explain how the following properties will vary down group 17 (Group VII) of the periodic table
A: Volatility
B: Electro-positivity

b. A solution of chlorine in water is used as a disinfectant. The following chemical equation indicates the reaction between water and chlorine

$$Cl_{2(g)} + H_2O(l) \rightarrow Cl^-(aq) + ClO^-(aq) + 2H^+(aq)$$

i. Give a balanced equation for the reaction that occurs when a concentrated solution of NaOH is added to this medium
ii. What name is given to this type of reaction?

c.

i. Write balanced equations for the preparation of HF and HI from CaF_2 and NaI respectively
ii. Explain any similarity or difference in the methods of preparation of HF and HI

d. Nitric acid is manufactured by catalytic oxidation of ammonia
i. Name the source of ammonia used in this process
ii. Write a balanced for the oxidation of ammonia to nitrogen monoxide
iii. Suggest the temperature at which the reaction in d(ii) occurred as well as the name of the catalyst that was used?
iv. Write an equation for the conversion of NO_2 to HNO_3

e. Sulphuric acid is prepared from Sulphur dioxide through the reaction

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

i. Why is a pressure of one atmosphere maintained taking into consideration that increasing the pressure will favor the production of more SO_3?
ii. Which catalyst is used in the process?